Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/08/2006
PI Name:	Marr, David W.M. Ph.D.		
Project Title:	A Novel Colloidal Microfluidics Platform for Spaceborne Micro Total Analysis Systems		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80401-1887	**Congressional District:**	7
Comments:	https://		
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	3
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG9-1364		
Performance Goal No.:			
Performance Goal Text:			

We use novel fabrication and actuation approaches for the creation of colloid-based microfluidic devices, including pumps, valves, and separators. To use novel fabrication and actuation approaches for the creation of colloid-based microfluidic devices. The “lab-on-a-chip” concept has gained significant research interest in recent years because the drastic reduction in length scales will allow tasks to be performed more rapidly, efficiently, and on smaller sample volumes than ever before. Functional systems fabricated to exploit microscale phenomena possess great promise to significantly streamline processes for fundamental research and medical applications in areas such as bioanalysis, medical diagnostics and therapeutics. Such developments will enable a large-scale shift from centralized laboratories to remote point-of-care and benchtop diagnostic facilities.
Seamlessly integrating individual devices capable of single operations is required however to finally bring to fruition the promise of µTAS (micro total analysis systems) as portable laboratories, chemical production facilities, remediation units, and health monitors. As human exploration and development of space extends from low Earth orbit and the

Task Description:	International Space Station to farther reaches of our solar system, both the need for sophisticated diagnostic equipment and the size and weight constraints placed on payload will surely increase. µTAS which are capable of both chemical analysis and isolating, sustaining and examining individual cells will be needed to meet these requirements. Generally, the utility, speed and performance of microfluidic chips increases as the overall device size decreases, particularly for devices that are ultimately designed for human implantation. The need to mix, administer and separate fluids at these length scales has long been a limiting factor in such devices. While some current microfluid handling devices and techniques enable functional devices at microscales, they may also impose significant constraints upon potential device capability, flexibility and performance. Our approach is to develop a microfluidics toolbox based on colloidal particles that will provide a small-scale integrated approach to microfluidics and the µTAS. Seamlessly integrating individual devices capable of single operations is required however to finally bring to fruition the promise of microTAS (micro total analysis systems) as portable laboratories, chemical production facilities, remediation units, and health monitors. As human exploration and development of space extends from low Earth orbit and the International Space Station to farther reaches of our solar system, both the need for sophisticated diagnostic equipment and the size and weight constraints placed on payload will surely increase. microTAS which are capable of both chemical analysis and isolating, sustaining and examining individual cells will be needed to meet these requirements. Generally, the utility, speed and performance of microfluidic chips increases as the overall device size decreases, particularly for devices that are ultimately designed for human implantation. The need to mix, administer and separate fluids at these length scales has long been a limiting factor in such devices. While some current microfluid handling devices and techniques enable functional devices at microscales, they may also impose significant constraints upon potential device capability, flexibility and performance. Our approach is to develop a microfluidics toolbox based on colloidal particles that will provide a small-scale integrated approach to microfluidics and the microTAS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	(Period covering 5/04-10/05) To date, we have made significant progress in our efforts to incorporate colloids as the functional units of microfluidic devices, having created microfluidic pumps, valves, and single particle/cell separators. To broaden their utility and application toward bioanalysis, current work has focused on the development of technologies for integrating these devices and their control hardware into portable platforms. These efforts have led not only to the use of capillary-based pumps that are self-driven and can be used to control the relative flow rates of differing streams during analysis (T. Stiles, R. Fallon, T. Vestad, J. Oakey, D.W.M. Marr, J. Squier, R. Jimenez, “Hydrodynamic Focusing for Vacuum-Pumped Microfluidics”, Microfluidics and Nanofluidics, v1, 280-283, 2005) but also used as the basis of fluid waveguiding techniques (M. Brown, T. Vestad, J. Oakey, D.W.M. Marr “Switchable Optical Waveguides via Microfluidic Flows”, in preparation for Applied Physics Letters). Also, to aid incorporation of optical control of separations we have continued development of easily integrated diode-based light sources (R.W. Applegate Jr., J. Squier, T. Vestad, J. Oakey, D.W.M. Marr, P. Bado “Microfluidic sorting system based on optical waveguide integration using femtosecond micromachining”, submitted to Optics Express).
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Other Journals or Periodicals	Vestad T, Brown M, Oakey J, Marr DWM. "Reconfigurable Microfluidic Waveguides for On-Chip Flow Cytometry." Micro Total Analysis Systems. 2004 Oct;1:653-5. , Oct-2004
Articles in Peer-reviewed Journals	Brown M, Vestad T, Oakey J, Marr DWM. "Optical waveguides via viscosity-mismatched microfluidic flows." Appl Phys Lett. 2006 Mar 27;88(13):134109 https://doi.org/10.1063/1.2190487 , Mar-2006
Articles in Peer-reviewed Journals	Santana-Solano J, Wu DT, Marr DW. "Direct measurement of colloidal particle rotation and field dependence in alternating current electrohydrodynamic flows." Langmuir. 2006 Jun 20;22(13):5932-6. PubMed PMID: 16768532 , Jun-2006
Articles in Peer-reviewed Journals	Applegate RW Jr, Squier J, Vestad T, Oakey J, Marr DWM, Bado P, Dugand MA, Said AA. "Microfluidic sorting system based on optical waveguide integration and diode laser bar trapping." Lab on a Chip. 2006;6:422-6. https://doi.org/10.1039/B512576F , Jan-2006
Articles in Peer-reviewed Journals	Stiles T, Fallon R, Vestad T, Oakey J, Marr DWM, Squier J, Jimenez R. "Hydrodynamic focusing for vacuum-pumped microfluidics." Microfluidics and Nanofluidics. 2005 Jul;1(3):280-3. https://doi.org/10.1007/s10404-005-0033-z , Jul-2005
Papers from Meeting Proceedings	Munakata T, Marr DWM. "Micro-/Nanofluidic Computing and its Applications to Biomedicine." 3rd IASTED International Conference on Biomedical Engineering, Innsbruck, Austria, February 2005. In: Proceedings of the 3rd IASTED International Conference on Biomedical Engineering, Innsbruck, Austria, 2005. , Feb-2005
Patents	6,802,489. granted October 2004. Oct-2004 Marr DWM, Gong T, Oakey J, Terray A. "Micro-fluidic valve with a colloidal particle element."
Presentation	D.W.M. Marr "Complex Fluid Manipulation with Applied Fields in Microfluidic Geometries" Physics Department, University of Colorado, Boulder, (invited). Mar-2005
Presentation	T. Munakata, D.W.M. Marr "Micro-/Nanofluidic Computing and its Applications to Biomedicine" 3rd IASTED International Conference on Biomedical Engineering, Innsbruck, Austria Feb-2005
Presentation	D.W.M. Marr "Colloids for Micro- and Nanotechnology" National Academy of Sciences, Frontiers of Science, Irvine, CA, (invited). Nov-2004

Presentation	D.W.M. Marr "Field-Based Colloidal Manipulation for Microfluidics and Photonics" Chemistry Department, Florida State University, Tallahassee, FL, (invited) Oct-2004
Presentation	T. Vestad, M. Brown, J. Oakey, D.W.M. Marr "Microfluidic Waveguiding for Micro Total Analysis Systems" muTAS 2004, Malmo, Sweden Sep-2004
Presentation	D.W.M. Marr "Field-Based Colloidal Manipulation for Microfluidics and Photonics" Chemical Engineering Department, University of Delaware, Newark, DE, (invited) Sep-2004
Presentation	T. Vestad, J. Oakey, D.W.M. Marr "Optical Trapping for Complex Fluid Microfluidics" 2004 SPIE conference on Optical Trapping, Denver, CO Aug-2004
Presentation	Marr, D.W.M. "Complex Fluid Manipulation with Applied Fields in Microfluidic Geometries" Physics Department, Universität Stuttgart, Stuttgart, Germany (invited) Jul-2005
Presentation	Marr, D.W.M. "Field-Based Colloidal Manipulation for Microtechnology" Chemical Engineering Department, University of Colorado, Boulder, Boulder, CO (invited) Oct-2005
Presentation	Marr, D.W.M. "Field-Based Colloidal Manipulation for Microtechnology" Chemical Engineering Department, Carnegie Mellon University, Pittsburgh, PA (invited) Oct-2005
Presentation	J. Oakey, T. Vestad, D.W.M. Marr, R. Applegate, J. Squier "Microfluidic Fluorescent Activated Cell Sorting With Optical Forces" µTAS 2005, Boston, MA Oct-2005